

Individual Contribution Statement

Assignment Title: Design and Development of an AI-Powered Smart University Assistant

Course: Artificial Intelligence / Machine Learning

This document outlines the specific contributions made by each group member in the design, development, and documentation of the AI-Powered Smart University Assistant. Both members collaborated actively throughout the project, ensuring all assignment requirements were met successfully.

Team Members

1. Izza Fatima
 2. Muhammad Habib
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1. Izza Fatima

Primary Responsibilities: Data Engineering, Recommendations, and Analysis

- **Problem Analysis & Dataset Understanding:** Led the initial analysis of the OULAD dataset, defined the problem statement, and finalized system requirements.
- **Data Preparation & Feature Engineering:** Cleaned the raw data, handled missing values, encoded categorical variables, and created advanced engineered features (e.g., active learning days, total VLE clicks).
- **Exploratory Data Analysis (EDA):** Produced comprehensive statistical summaries and created meaningful visualizations comparing successful and at-risk students.
- **Recommendation System:** Designed and implemented the personalized learning resource recommendation module (`recommender.py`), mapping student risk levels and weaknesses to targeted resources.
- **Documentation & Reporting:** Authored major sections of the final project report (`final_report.pdf`), including dataset description, data preprocessing, and exploratory analytics.

2. Muhammad Habib

Primary Responsibilities: Machine Learning, NLP, and Application Development

- **Machine Learning Model Development:** Trained and evaluated multiple classification models (Logistic Regression, Random Forest, XGBoost) and developed the early-warning prediction module (`prediction.py`).
- **University FAQ Chatbot:** Developed the NLP-based question-answering chatbot (`chatbot.py`), including text preprocessing, semantic similarity calculation, and fallback mechanisms.
- **Application & Dashboard Development:** Designed and built the interactive Streamlit web application (`app.py`), integrating all underlying modules into a cohesive, user-friendly interface.
- **Explainable AI (XAI):** Implemented global and local model explainability using SHAP (`explainability.py`) to provide transparent reasoning for risk predictions.
- **Ethics & System Testing:** Analyzed the ethical and privacy implications of the system, and performed extensive testing across the application pages to ensure functionality.

Joint Contributions

- Architecture design and system workflow integration.
- Recording of the final 5-7 minute demonstration video.
- Proofreading and finalization of the project documentation and README file.

We hereby confirm that the work submitted is original and the contributions stated above accurately reflect each member's involvement.